

PROOF-CARRYING REDUCTIONS FOR 4-REGULAR GIRTH-NINE GRAPHS OF EXCESS FOUR

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ABSTRACT. The first open order in the quartic girth-nine cage problem is 165, four vertices above the Moore bound. We derive a distance-five commutation identity for any hypothetical 4-regular graph of girth at least nine on 165 vertices and use it to constrain the distribution of its 9-cycles. If c_v is the number of edges induced by the four vertices outside the radius-four Moore tree rooted at v , then

$$\sum_v c_v \equiv 6 \pmod{9}, \quad \sum_v c_v \geq 15.$$

In particular, such a graph would contain at least 2825 cycles of length nine. In the equality case, the odd part of a naturally edge-weighted support pair (K, q) is a single cycle of length 9, 11, 13, or 15; here K is always the unweighted support graph. We exclude length 13 by three independently replayed LRAT certificates. For length 11, we give two independent complete orbit enumerations: exactly 24 candidate weighted support pairs pass the elementary constraints, up to the dihedral action and relabelling of outside vertices. Seventeen fail a locally closed commutation system, again with checked LRAT certificates. Distance-five separation excludes three of the remaining seven, leaving four explicit local witnesses. Thus the equality case is reduced to the length-9 and length-15 branches and four explicit length-11 supports. These are strict necessary conditions and proof-carrying finite reductions; they do not assert the nonexistence of order 165.

At the rooted global level, we independently enumerate 134 nonempty outer-data orbits. We also classify the perfect matchings forced by a zero branch-pair deficit. There are exactly 6^{27} labelled admissible matchings. Rooted-tree automorphisms reduce them to $6^4 = 1296$ normalized S_3 -gain matrices and exactly 21 double-action classes. Two independent enumerators agree on the classes and their sizes. For the first outer orbit, these representatives give a lossless cover by 21 full-completion shards. Within every shard, a fixed matching edge pair and two still-variable edges between another branch pair expose 2916 distinct forbidden 8-cycles. We serialize the resulting negative binary clauses and verify all 61,236 cycle witnesses by an independent reconstruction. This is an exact global reduction, not a decision of any shard or of order 165.

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1. INTRODUCTION

A (k, g) -graph is a finite simple k -regular graph of girth at least g , and $n(k, g)$ denotes the least possible order. The cage problem asks for $n(k, g)$ and extremal constructions; see the dynamic survey [3]. For $(k, g) = (4, 9)$, the Moore bound is

$$1 + 4 + 4 \cdot 3 + 4 \cdot 3^2 + 4 \cdot 3^3 = 161.$$

The current certified lower bound is 165 [2], while the smallest currently reported construction has order 270 [4]. Consequently, order 165 is both the first open order and the first possible excess-four case.

Our purpose is not to report a new lower bound. Instead, we isolate rigid necessary structure at order 165 and make every finite exclusion independently checkable. The mathematical reduction has three levels. First, the graph at distance five from a vertex is again 4-regular and its adjacency matrix is a polynomial in the original adjacency matrix, modulo the all-ones matrix. Second, the resulting commutation identities define a small unweighted support graph K together with positive integral edge weights q in the extremal mass case. Exact orbit enumeration and proof-producing SAT then reduce the pair (K, q) to a short list. Third, at the rooted global level, the two-edge girth constraints turn every zero-deficit branch pair into an antidistance perfect matching between two rooted ternary trees. A gain-matrix normal form reduces all such matchings to 21 types. Fixing one type then turns a complete family of four-completion-edge 8-cycles into binary clauses.

The proof artifacts use DIMACS formulas and textual LRAT proofs. LRAT is a hinted clausal proof format designed for simple certified replay [1]; the formulas were solved with CaDiCaL [5]. Our independent verifier reconstructs each clause from the mathematical definitions before replaying the LRAT hints. Thus a solver log, a timeout, or a numerical observation is never used as a proof.

2. THE MOORE-TREE COMPLEMENT AND DISTANCE FIVE

Let G be a 4-regular graph on 165 vertices with girth at least nine, and let A be its adjacency matrix. Fix $v \in V(G)$. The radius-four ball about v is the Moore tree, with level sizes

$$1, 4, 12, 36, 108.$$

Write $X(v)$ for the four vertices outside this tree and set

$$c_v = |E(G[X(v)])|.$$

The graph $G[X(v)]$ is a forest. Every vertex of this forest has internal degree at most three, hence has a neighbour in the fourth Moore level. Thus every member of $X(v)$ is at distance exactly five from v .

Let H be the distance-five graph of G , and also write H for its adjacency matrix. Each vertex has exactly four distance-five neighbours, so H is a simple 4-regular graph. The tree recurrences through distance four give

$$\begin{aligned} A_2 &= A^2 - 4I, & A_3 &= A^3 - 7A, \\ A_4 &= A^4 - 10A^2 + 12I. \end{aligned}$$

As every pair of vertices has a unique distance in $\{0, 1, 2, 3, 4, 5\}$,

$$(1) \quad H = J - P(A), \quad P(x) = x^4 + x^3 - 9x^2 - 6x + 9.$$

Regularity gives $AJ = JA$, hence

$$(2) \quad AH = HA.$$

For a G -edge uv , define

$$q_{uv} = |N_H(u) \cap N_H(v)| = (H^2)_{uv}.$$

Let K be the unweighted simple graph on $V(G)$ whose edges are precisely the G -edges with $q_{uv} > 0$, and write K also for its 0–1 adjacency matrix. Thus (K, q) is an edge-weighted support pair; the matrix K itself is never q -weighted. If $uv \in E(H)$, every summand contributing to $(AH)_{uv}$ or $(HA)_{uv}$ comes from a positive- q G -edge. Consequently (2) restricts on H -edges to the local identities

$$(3) \quad (KH)_{uv} = (HK)_{uv} \quad (uv \in E(H)).$$

Taking the (v, v) entry after multiplying (2) by H gives

$$(4) \quad \sum_{u \sim_G v} q_{uv} = 2c_v.$$

Indeed, $(HAH)_{vv}$ counts the two orientations of the G -edges induced by $N_H(v) = X(v)$. Summation over v yields

$$(5) \quad S := \sum_v c_v = \sum_{uv \in E(G)} q_{uv}.$$

3. A GLOBAL CYCLE-COUNT REFINEMENT

The rooted Moore-tree count gives exactly $154 + c_v$ cycles of length nine through v . If $C_9(G)$ denotes the number of distinct 9-cycles, then

$$(6) \quad 9C_9(G) = \sum_v (154 + c_v) = 165 \cdot 154 + S.$$

In particular,

$$(7) \quad S \equiv 6 \pmod{9}.$$

Theorem 3.1. *Every 4-regular graph of girth at least nine on 165 vertices satisfies*

$$S \geq 15, \quad C_9(G) \geq 2825.$$

Proof. By (7), the only smaller nonnegative possibility is $S = 6$. Let $R = \{v : c_v > 0\}$. Equation (4) shows that R is exactly the set of vertices incident with K . Since $|R| \leq 6$, the girth condition makes $K[R]$ a forest.

On a forest, positive integral edge weights whose weighted degree is even at every vertex are all even: remove a leaf and continue inductively. Write $q_e = 2w_e$. Then (5) gives $\sum_e w_e = 3$. Moreover, for every active vertex v ,

$$d_K(v) \leq c_v = \sum_{e \ni v} w_e \leq |E(K)| - d_K(v).$$

The right inequality follows because an edge incident with v cannot be an edge induced by $N_H(v)$. It follows that K has exactly three edges, every active degree is one, and every w_e equals one. Thus K is a perfect matching on six vertices with $q_e = 2$ on its edges.

It remains to enumerate $H[R]$, disjoint from this fixed matching, with maximum degree four, the prescribed common-neighbour counts, the diagonal identities (4), and the local identities (3) supported on $H[R]$. There are $2^{12} = 4096$ raw edge masks. Exactly 120 pass all conditions before commutation; they form one coloured isomorphism type, in which $H[R]$ is the complement of a 6-cycle. Every representative fails (3) on an H -edge. The edge weights q occur only in the common-neighbour and diagonal constraints. The direct enumerator and an independently parameterized verifier both return zero survivors. Hence $S \neq 6$, and the two inequalities follow from (7) and (6). $\square$

4. THE EQUALITY SUPPORT

Assume from now on that $S = 15$. Reduce (4) modulo two and let K_{odd} consist of the edges of K having odd weight. Every vertex has even degree in K_{odd} , so this graph is Eulerian. It is nonempty, has an odd number of edges, and has at most 15 edges. The girth condition prevents two edge-disjoint cycles, since they would use at least 18 edges. Therefore K_{odd} is a single cycle of length

$$(8) \quad 9, 11, 13, \text{ or } 15.$$

Theorem 4.1. *Under $S = 15$, the odd support K_{odd} is not a 13-cycle.*

Proof. The unit 13-cycle uses weight 13, leaving mass two. Up to dihedral symmetry, there are exactly three possibilities:

- (i) one cycle edge has weight three;
- (ii) there is one additional weight-two edge with exactly one endpoint on the cycle;
- (iii) there is one isolated weight-two edge.

An additional chord of the 13-cycle would form a cycle of length at most seven. For each of the three supports, Boolean variables encode $H[R]$ and the locally closed conditions listed in section 6. The exact formula sizes are shown in table 1. Each formula is UNSAT and its textual LRAT proof passes independent replay. $\square$

TABLE 1. Certified exclusion of the three length-13 supports.

case	$ R $	variables	clauses	LRAT additions
one weight-three cycle edge	13	182	5,372	525
pendant weight-two edge	14	215	9,095	316
isolated weight-two edge	15	247	15,797	127

5. COMPLETE CLASSIFICATION FOR THE LENGTH-ELEVEN BRANCH

Only four units of mass remain above a unit 11-cycle. Since H is 4-regular, every q_{uv} is at most four. A cycle edge therefore has weight one or three, and a positive noncycle edge has weight two or four. Thus there are four mass partitions: two promoted cycle edges; one promoted cycle edge and one weight-two noncycle edge; one weight-four noncycle edge; or two weight-two noncycle edges.

Theorem 5.1. *Suppose $S = 15$ and $K_{\text{odd}} = C_{11}$. Up to the dihedral group of the cycle and arbitrary relabelling of the at most four additional active vertices, there are exactly 24 candidate weighted support pairs (K, q) satisfying the elementary degree, parity, and girth constraints. Their family counts are as follows.*

mass partition	orbits
two promoted cycle edges	5
one promoted cycle edge and one weight-two edge	7
one weight-four edge	2
two weight-two edges	10
total	24

Of these 24 supports, 17 violate the locally closed commutation system. Of the other seven, distance-five separation excludes three more. Exactly four supports survive the strengthened local system and admit explicit local witnesses.

Proof. The first enumerator varies the number of outside active vertices from zero through four, tests all 16,528 placements of the four remaining mass units, and applies the following exact filters: support degree at most four; positive even weighted degree at most six; and support girth at least nine. It finds 381 active-labelled placements and canonicalizes them to 24 orbits.

The independent enumerator instead fixes a universe consisting of the 11-cycle and four optional outside vertices. It permits inactive outside vertices and quotients only at the end by $D_{11} \times S_4$, a group of order 528. It checks 5,554 fixed-universe placements, accepts 1,660, and reconstructs exactly the same set of 24 canonical codes.

For each support, we then generate the locally closed formula described in [section 6](#). Seventeen formulas have independently replayed LRAT refutations. The other seven initially have explicit local witnesses.

It remains to use the meaning of H : its edges join vertices at distance exactly five in G . Since every edge of K is an edge of G , an H -edge uv is impossible whenever u and v are joined in K by a path of length two, three, or four. Adding precisely these unit clauses makes the formulas for 111_02, 111_03, and 111_09 UNSAT; their LRAT proofs pass an independent replay. The formulas for 111_05, 111_07, 111_15, and 111_18 remain SAT, and a separate verifier checks the four explicit edge sets against every equation and every distance-separation clause. This proves both assertions without treating nonterminal solver output as evidence. $\square$

The four surviving support identifiers in the canonical catalogue are

111_05, 111_07, 111_15, 111_18.

They are local witnesses only, not 165-vertex graphs.

Corollary 5.2. *If a 4-regular graph of girth at least nine and order 165 satisfies $S = 15$, then its odd support is either a 9-cycle, a 15-cycle, or an 11-cycle whose weighted support pair (K, q) is one of the four explicit survivors above.*

6. CERTIFICATE MODEL AND INDEPENDENT AUDIT

Fix a weighted support pair (K, q) on its active set R , where K is the unweighted simple support graph (and also its 0–1 adjacency matrix) and $q : E(K) \rightarrow \mathbb{Z}_{>0}$ is the edge-weight function. Boolean variables encode all possible edges of $H[R]$ outside K . The CNF enforces:

- (C1) $H[R]$ is disjoint from K and has maximum degree four;
- (C2) every K -edge uv has exactly its displayed number q_{uv} of common $H[R]$ -neighbours;
- (C3) for every $r \in R$, the number of K -edges induced by $N_{H[R]}(r)$ is half the weighted K -degree of r ;
- (C4) conditional on every possible $H[R]$ -edge uv , the entry $(KH)_{uv} = (HK)_{uv}$ from (3) holds, with K unweighted;
- (C5) if the distance from u to v in the unweighted support K lies in $\{2, 3, 4\}$, then $uv \notin E(H[R])$.

Products use three-clause Boolean AND equivalences. Cardinality conditions use direct subset clauses, and conditional commutation uses complete truth tables.

These restrictions are exact necessary conditions on the active set. For example, if $uv \in E(H[R])$ and a vertex $y \notin R$ contributed to $(AH)_{uv}$, then uy would be a G -edge and u, y would have the common H -neighbour v . Hence $q_{uy} > 0$, contradicting $y \notin R$. The same argument applies to $(HA)_{uv}$. All common neighbours counted on a positive- q edge likewise belong to R .

The audit has three independent layers.

- (1) A second orbit enumerator reconstructs the length-11 catalogue from a different fixed-universe parameterization.
- (2) The certificate verifier regenerates every intended clause, including each distance-separation unit, from the weighted pair (K, q) and requires literal equality with the DIMACS formula.
- (3) A standard-library LRAT replay checks every hinted propagation step up to the empty clause; it does not invoke CaDiCaL or any other SAT solver. SAT cases are checked by direct substitution instead.

There are no random seeds, floating-point comparisons, heuristic cut generators, or timeouts in the certified conclusions.

7. ANTIDISTANCE MATCHINGS IN THE ROOTED COMPLETION

By [theorem 3.1](#), some root r has $c_r > 0$. The 108 leaves of its radius-four Moore tree split into four sets $L_0, \dots, L_3$ of size 27. For $i < j$, put

$$\delta_{ij} = 27 - |E(L_i, L_j)|.$$

A leaf has at most one completion neighbour in any other fixed branch: two such edges, together with the path of length at most six inside that branch, would form a cycle of length at most eight. Hence $\delta_{ij} = 0$ forces $E(L_i, L_j)$ to be a perfect matching.

Label the leaves in either rooted branch by triples in $\{0, 1, 2\}^3$ and write $\ell(x, y)$ for the length of the longest common prefix of two distinct triples. Their tree distance inside the branch is $6 - 2\ell(x, y)$.

Theorem 7.1 (Antidistance matching theorem). *Let $f : \{0, 1, 2\}^3 \rightarrow \{0, 1, 2\}^3$ be a bijection. The perfect matching $x \mapsto f(x)$ creates no cycle of length at most eight together with the two rooted ternary trees if and only if*

$$(9) \quad \ell(x, y) + \ell(f(x), f(y)) \leq 2 \quad (x \neq y).$$

There are exactly 6^{27} such labelled matchings. Modulo independent automorphisms of the source and target rooted trees, they form exactly 21 classes.

Proof. The cycle determined by two matching edges $xf(x)$ and $yf(y)$ has length

$$2 + (6 - 2\ell(x, y)) + (6 - 2\ell(f(x), f(y))).$$

It is even, so requiring length at least nine is equivalent to (9).

Write $x = (a, b, c)$ and $f(x) = (A, B, C)$. Condition (9) is equivalent to the following three elementary conditions:

- (a) for fixed (a, b) , the three values of A are distinct;
- (b) for fixed a , the nine pairs (A, B) are distinct;
- (c) for fixed (A, B) , the three values of C are distinct.

The forward implication follows by considering input common-prefix lengths two and one; condition (c) then follows from bijectivity. Conversely, (a) handles input prefix length two, (b) handles prefix length one, while (b) and (c) together make all 27 outputs distinct and handle prefix length zero.

For each fixed a , choose the three permutations $c \mapsto A$, one for each b , in 6^3 ways. For each fixed A , the three resulting inputs must receive distinct values of B , giving another 6^3 choices. This is 6^6 choices for each a . Finally, for each of the nine output prefixes (A, B) , condition (c) gives 6 choices for C . The labelled count is therefore

$$(6^6)^3 6^9 = 6^{27}.$$

Let W be the automorphism group of a rooted ternary tree of depth three; $|W| = 6^{13}$. Its nine bottom-node permutations normalize $A = c$, and the nine bottom-node permutations on the target normalize $C = a$. Every matching is consequently equivalent to

$$(10) \quad f_P(a, b, c) = (c, P_{a,c}(b), a),$$

where $P = (P_{a,c})$ is a 3-by-3 matrix over S_3 .

To see exactly which normal forms remain equivalent, write source and target tree automorphisms in wreath coordinates. Comparing the first and third coordinates in an equivalence $kf_P = f_Qh$ forces the source bottom permutations to be one common column permutation ρ , and the target bottom permutations to be one common row permutation α . The middle coordinate is then precisely

$$(11) \quad Q_{\alpha(a), \rho(c)} = \gamma_c P_{a,c} \beta_a^{-1}$$

for row and column switches $\beta_a, \gamma_c \in S_3$. Conversely every transformation in (11) comes from rooted-tree automorphisms.

Switching makes the first row and column of P the identity. Explicitly, after permuting rows and columns, the normalized entries are

$$(12) \quad N_{a,c} = P_{0,0}^{-1} P_{a,c} P_{a,0}^{-1},$$

up to simultaneous conjugation. Only the four entries $N_{1,1}, N_{1,2}, N_{2,1}, N_{2,2}$ remain, so the normalized state space has $6^4 = 1296$ elements.

The final finite quotient was computed twice. For each normalized state, the first program forms all $6^3 = 216$ one-step images obtained from a row permutation, a column permutation, switching normalization (12), and simultaneous conjugation. It treats these images only as edges of an equivalence graph and takes transitive closure; the 216 transformations are not assumed to form a group. The second program uses only six elementary transposition moves and an independent union-find traversal. It additionally checks that every representative reported by the first program lies in a different component and that its recorded class size equals that component's size. Both therefore obtain the same 21 equivalence-closure components representative by representative. Their component-size distribution is

size	1	2	6	12	18	27	36	54	108	216
multiplicity	1	1	2	1	2	1	2	3	7	1

whose weighted sum is 1296. Both enumerations use exact permutations and exhaust the displayed finite state space. $\square$

The rooted outer data can also be quotiented before any completion search. For the five possible nonempty forests induced by the four exterior vertices, two independent enumerators give respectively

$$8, 22, 25, 63, 16$$

orbits of attachment masks and branch-pair deficits, for a total of 134. The following corollary records how [theorem 7.1](#) refines the first of these global orbits.

Corollary 7.2 (Lossless 21-shard cover). *Consider the rooted outer orbit with exterior forest $K_2 + 2K_1$, attachment masks $(7, 7, 15, 15)$, and deficit vector*

$$(\delta_{01}, \delta_{02}, \delta_{03}, \delta_{12}, \delta_{13}, \delta_{23}) = (1, 1, 2, 3, 0, 0).$$

Its full graph completions are the union of 21 lossless shards obtained by fixing the $(1, 3)$ perfect matching to the representatives in [theorem 7.1](#).

Proof. The product $W_0 \times W_1 \times W_2 \times W_3$ of the four rooted-branch automorphism groups preserves the exterior forest, masks, deficits, degrees, and cycle lengths. Use $W_1 \times W_3$ to send the $(1, 3)$ matching to one of the 21 representatives. This relabels all attachments and all other horizontal edges equivariantly, but they remain unrestricted variables. The old normalization of the x_0 attachment in branch 1 must therefore be released. The corresponding normalizations in branches 0 and 2 use the commuting factor $W_0 \times W_2$ and may be retained. After fixing a matching representative, no quotient by its joint stabilizer is imposed; all remaining variables are left available. Thus no completion is lost, and every object in a shard is a completion in the stated outer orbit.

The independent audit checks this construction at clause level. It removes exactly the 27 signed units belonging to the branch-1 attachment gauge, changes no other base clause, and appends 729 signed units in each shard: 27 select the matching and 702 exclude the other branch-pair edges. It also verifies that the 21 representatives lie in 21 distinct components of the independent 1296-state enumeration, with the recorded component sizes. This proves the cover without assuming any outcome of a completion solver. $\square$

Theorem 7.3 (Fixed-matching four-edge cuts). *Fix one of the branch- $(1, 3)$ perfect matchings f in [theorem 7.2](#). Let $a, b \in L_1$ be sibling leaves and write $x = f(a)$ and $y = f(b)$. If u, v are sibling leaves in L_0 or in L_2 , then neither*

$$\{xu, yv\} \quad \text{nor} \quad \{xv, yu\}$$

can be contained in a completion of girth at least nine. These obstructions give exactly 2916 distinct negative binary clauses for every one of the 21 matching shards.

Proof. The paths from a to b and from u to v in their respective rooted ternary trees both have length two. Suppose first that xu and yv are selected. The two fixed matching edges ax, by , these two selected edges, and the two tree paths form a cycle of length

$$2 + 2 + 4 = 8.$$

Its vertices are distinct: a, b lie in L_1 , x, y lie in L_3 , and u, v lie in L_0 or L_2 ; within each pair the vertices differ. Thus the cycle is simple. Interchanging u and v proves the other exclusion.

There are 9 bottom sibling triples and three unordered pairs in each, hence 27 sibling pairs in every branch. We may choose the input pair in 27 ways, the target pair in $27 + 27$ ways, and one of two orientations. This gives

$$27(27 + 27)2 = 2916.$$

The clauses are distinct because their two L_3 endpoints recover x, y , the fixed bijection recovers the unordered input pair, and their endpoints in L_0 or L_2 recover the target pair and orientation. $\square$

Corollary 7.4. *Appending the 2916 clauses from [theorem 7.3](#) to each shard of [theorem 7.2](#) is lossless with respect to genuine girth-at-least-nine completions. Across all 21 representatives this gives 61,236 serialized simple-cycle certificates.*

Proof. Every appended clause excludes only the simultaneous selection of the two free edges in one of the simple 8-cycles from [theorem 7.3](#). Hence no genuine completion is removed. The independent certificate program rebuilds the global candidate-variable order, every normalized matching, and every sibling pair without importing the generator. It reconstructs exactly 2916 clauses in each of the 21 files and checks an explicit simple 8-cycle witness for every clause. A second byte-level audit verifies that each materialized formula consists of the unchanged matching-shard prefix followed by exactly its certified binary-clause suffix. $\square$

8. LIMITATIONS AND NEXT STEPS

The conclusions are deliberately conditional. They neither construct an order-165 graph nor exclude one, and therefore do not improve the interval $165 \leq n(4, 9) \leq 270$. The length-9 branch and the global completion of the four length-11 survivors remain open. A directly checked strengthened local witness also survives for the length-15 branch: its $H[R]$ is 4-regular on all 15 active vertices, and every one of its edges has K -distance at least five. Thus the present active-set equations, including distance separation, cannot exclude this branch.

The 134 rooted outer orbits form a complete global gate for order 165, and [theorems 7.2](#) and [7.4](#) subdivide and strengthen one of them without loss. Neither that outer orbit nor any of its 21 matching shards is decided here. A global nonexistence proof would require a complete checked exclusion of all

134 outer orbits; the present classification is a structural reduction toward that objective, not such an exclusion.

DATA AND CODE AVAILABILITY

The computational supplement accompanying this manuscript contains the two support-orbit enumerators, all canonical support records, the CNF generator, the exact SAT witnesses, textual LRAT certificates, and independent replay programs. It also contains both antidistance-matching enumerators, the full list of 21 representatives, the independent representative-by-component audit, the lossless shard generator and audit, and the 61,236 fixed-matching four-edge cycle certificates with an independent reconstruction and materialization audit. Every table and finite claim in the paper can be regenerated using the commands in the package’s run log.

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